

Problem Statement

XYZ Retail Corp sells products across multiple categories and regions in the United States. Management wants to understand where their sales are concentrated — by region, category, customer segment, and time period — in order to plan inventory, marketing, and expansion strategy more effectively. As a business analyst, you are required to analyze the company's order-level sales data using Power BI and build an interactive dashboard that reveals sales trends, top-performing products/customers, and regional/segment-wise demand patterns.

Objective

1. Import the dataset into Power BI
2. Design and build an interactive dashboard that answers the business questions below.
3. Add the missing reports
4. Organize the dashboard properly

Academic Integrity & Submission Guidelines

1. This assignment will be allocated zero marks for any kind of malpractice or use of AI tools/other unauthorized solutions to complete the work.
2. In case of copying or sharing of work, both the student sharing and the student consuming the work will be penalized equally.
3. Students involved in malpractice will not be allowed to take up the next Assignment/CIA/Project.
4. Any student involved in malpractice will be asked to meet the HOD.
5. Questions will be asked in class to evaluate the authenticity and understanding of your work. Any student who fails to explain their own dashboard/logic will be allocated zero marks.
6. No two submissions should be identical in dashboard design, or written insights — even if working with the same dataset, analysis and presentation must be independent.
7. Submissions must include the student's own screenshots of the working Power BI file (with student name & roll number visible in the file name and report title), not just a PDF export.
8. Students must be able to explain every measure, calculated column, and visual choice used in their dashboard during the viva.
9. Late submissions will not be accepted

10. The .pbix file must be submitted along with any PDF, so originality and formula logic can be verified.
11. Use of pre-built dashboard templates downloaded from the internet is considered malpractice; the dashboard must be built from scratch by the student.
12. Plagiarized or AI-generated written insights/recommendations (copy-pasted without genuine interpretation) will be treated the same as dashboard-level malpractice.

Dashboard Structure

- KPI cards: Total Sales, Total Orders, Total Customers, Avg. Sales per Order
- Sales trend over time (by month/quarter/year, using Order Date)
- Sales by Region/State
- Sales by Category & Sub-Category
- Top 10 best-selling products by Sales
- Top 10 customers by Sales, with Segment breakdown

Questions

1. Which categories and sub-categories generate the highest sales?
2. Which regions/states/cities are the top revenue contributors?
3. Who are the top 10 customers by sales value, and which segment do most high-value customers belong to?
4. What is the month-over-month or year-over-year sales trend? Are there any seasonal spikes or drops?
5. Based on your dashboard, what are 2 actionable recommendations you would give management regarding inventory or regional focus?